

PUBLISHED

Question	Answer	Marks	Guidance
6(a)	H_0 : population median mark for school A is equal to population median mark for school B . H_1 : population median mark for school A is not equal to population median mark for school B .	B1	Allow m , but not μ . If m defined, must see population .
	$\frac{1}{2} \times 123(123 + 147 + 1)$ [=16666.5] $\frac{1}{12} \times 123 \times 147(123 + 147 + 1)$ [=408329.25]	B1	Both expressions correct. Implied by 16666.5 and 408329.25 seen.
	$\frac{15355 + 0.5 - '16666.5'}{\sqrt{408329.25}}$	M1	OR attempts to find critical value or p -value using <i>their</i> mean and variance. Condone continuity correction omitted or incorrect.
	-2.05	A1	AWRT -2.05 or CV = 15414 or $p = 0.0201$. Condone continuity correction omitted or incorrect.
	-2.05 < -1.96 OR 0.980 > 0.975 OR 0.020 < 0.025 OR 15355 + 0.5 < 15414, reject H_0 / significant.	M1	Compare <i>their</i> ± 2.05 with ± 1.96 (consistent signs) or <i>their</i> 0.980 with 0.975, or <i>their</i> 0.020/0.040 with 0.025/0.05 or <i>their</i> 15355.5 with 15414 (condone continuity correction omitted or incorrect). Consistent signs and appropriate result (may be in terms of H_1). Test result can be implied by an attempt at an appropriate conclusion in context.
	Sufficient evidence to suggest that there is a difference in average marks between the students in school A and B . OR Sufficient evidence to suggest that, on average, the test scores for students at the two schools are not equal.	A1	Correct conclusion from correct working, ignoring hypotheses. Must consider continuity correction e.g. 15355.5 seen. In context. Level of uncertainty in language is used (for example, not 'prove'). Do not accept statements such as 'there is insufficient evidence to suggest...'.
		6	
6(b)	The (underlying) distributions are symmetrical/similar shape. OR The populations are symmetrical/similar shape. OR No repeated marks.	B1	
		1	